



DO-IT-YOURSELF: PROJECT

scribed using `sphinx`, which works great for WAV file transcription on Pi Zero.

```
C> Google STT : Google Speech-to-text....
```

Although not explicitly stated, this is far more efficient considering the limited capacity of Raspberry Pi Zero. It works in two ways—listens and transcribes, listens and records transcriptions. For listening and transcribing [in the gaps of speeches], it works great for Raspberry Pi 4 or Pi 5, but for Raspberry Pi Zero, the second option works great—record it for a duration, then stop and transcribe. To install Google STT, the `pip` module must be installed.

Necessary software for Google STT.

```
$pip install SpeechRecognition
```

Project code.

A few codes are provided here with descriptions:

1. `sr1.py` - live speech transcription using Google STT, works great for powerful computers

2. `sr2.py` - live speech transcription using PocketSphinx, works great for powerful computers

3. `sr3.py` - offline speech transcription. Uses WAV file and writes to `*.txt` file. Uses GTTS. It works great for Raspberry Pi Zero.

4. `sr4.py` - sets time frame, records live voice, and then transcribes to an `audio.txt` file. Works great for Raspberry Pi Zero.

Study this code carefully to understand where to change the language choice (for instance, conversion from Japanese speech to English text).

```
$> aplay your-file.wav
```

```
#to playback your recorded voice
```

```
$> arecord -D plughw:1,0 -f
```

```
S16_LE -r 16000 -d 10 audio.wav
```

```
#record your voice for 10 seconds at
```

16 bit and for 10 seconds.

Video link of the speech transcription process: <https://youtu.be/u9OHj92fPvQ>

Testing

After interconnecting the device based on Fig. 2 and using the cables shown in Fig. 3, power the device and run the project code for live transcription (`sr1.py`) and start speaking in English, and it will start transcribing.

For offline audio, run `sr3.py`. The transcribed audio text file can be found as `audio.txt`. Open this file to see the transcribed text. Fig. 4 shows a transcribed audio snippet.

Speech-to-text conversion is a serious business. Media houses use costly software and powerful computers to do it online during live telecasting. Doing it on Raspberry Pi Zero may seem like child's play, but there are other advantages to using these small power (5V only) Pi Zero computers. Some typical uses recommended for this Pi Zero speech transcription include:

1. An on-the-go voice assistant for hearing-impaired persons who can promptly decipher sentences while on the move.

2. Traveller's assistant in a foreign land, such as talking to a cab driver or waiter.

3. Robotics - For instance, selling voice-operated robot toys to Chinese buyers. Besides, consuming small power, it efficiently converts foreign speeches to English commands for further processing.

4. Handheld transcribers to be used at points-of-sale in supermarkets, quickly helping to understand foreign phrases used by buyers or sellers at that point. **EFY**



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“

Branding is what people say about you when you are not in the room.

”



-Jeff Bezos



EFY Note

The source code of this project is available

for free download at www.electronicsforu.com